

$\Rightarrow$ line integrals of vector fields expressed as
gradients of scalar functions are PATH-INDEPENDENT

$\Downarrow$

CONSERVATIVE FIELDS $\Leftrightarrow \vec{U} = \nabla f$

$\nabla f \cdot \hat{s} = \frac{df}{ds}$

CIRCULATION of a CONSERVATIVE FIELD

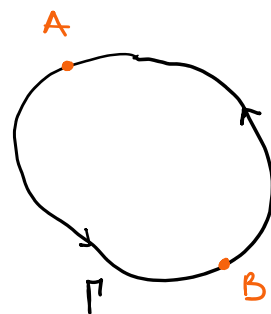
$$\oint_{\Gamma} \vec{U} \cdot d\vec{\ell} = \oint_{\Gamma} \boxed{\nabla f \cdot \hat{\ell}} d\ell = \oint_{\Gamma} \left(\frac{df}{d\ell} \right) d\ell =$$

$\downarrow$ $\downarrow$

$$= \int_A^B df + \int_B^A df = \boxed{0 \quad \forall \text{ CLOSED PATHS}}$$

$\downarrow$ $\downarrow$

$$f(B) - f(A) \quad f(A) - f(B)$$



• every conservative field is CIRCULATION-FREE

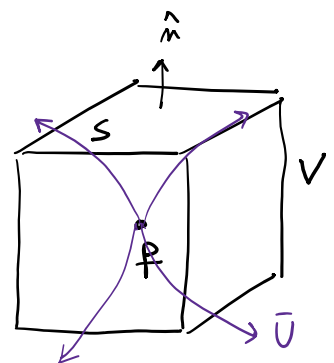
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DIVERGENCE $\mathbb{R}^3 \Rightarrow \mathbb{R}$

let $\vec{U}$ vector field on domain D $\hat{n} dS$

Def : $\nabla \cdot \vec{U}(P) = \lim_{\Delta V \rightarrow 0} \frac{\oint_{S_V} \vec{U} \cdot d\vec{S}}{\Delta V} = \left[\frac{\text{FLUX}}{\text{VOLUME}} \right]$

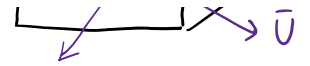
VOLUME of V Surface bounding V



VOLUME of V

Surface bounding V

local flux density



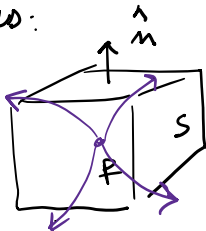
Flux of V.F. = $\int_S \vec{U} \cdot \hat{n} dS$ = ACCUMULATION of normal component of VF through a surface

↑
TAKE the component of $\vec{U}$ normal to S

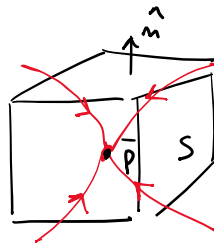
open / closed

CART. COORDS: $\nabla \cdot \vec{U}(\underline{r}) = \frac{\partial U(\underline{r})}{\partial x} + \frac{\partial U(\underline{r})}{\partial y} + \frac{\partial U(\underline{r})}{\partial z}$

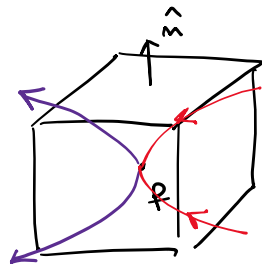
Examples:



$\nabla \cdot \vec{U}(\underline{r}) > 0$



$\nabla \cdot \vec{U}(\underline{r}) < 0$



$\nabla \cdot \vec{U}(\underline{r}) = 0$

DIVERGENCE theorem (GAUSS theorem)

let $\vec{U} \in C^1$ on D

↙ $\vec{U}$ is continuous
↘ first partial derivatives of $\vec{U}$ are continuous



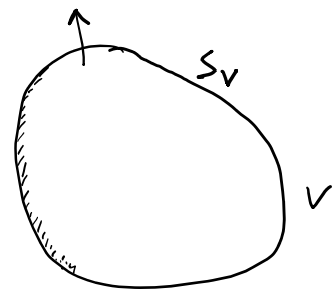
continuous

$\vec{U}$ are continuous

$$\int_V \underbrace{\nabla \cdot \vec{U}}_{\text{local flux density}} dV = \oint_{S_V} \underbrace{\vec{U} \cdot d\vec{S}}_{\text{normal component of } \vec{U} \text{ through } S_V}$$

accumulation of local flux density on V

accumulation of normal component of $\vec{U}$ through S_V



SOLENOIDAL FIELDS

$$\vec{U} \text{ is SOL. if } \oint_{S_V} \vec{U} \cdot d\vec{S} = 0 \quad \forall \text{ closed } S_V \in D$$

If: $\vec{U}$ is sol.

$$\oint_{S_V} \vec{U} \cdot d\vec{S} = 0$$

(Gauss)

$$\int_V \nabla \cdot \vec{U} dV = 0$$

$\Rightarrow$ if $\vec{U}$ is SOLENOIDAL



$$\nabla \cdot \vec{U} = 0 \quad \forall p \in D$$

$\vec{U}$ is DIVERGENCE-FREE

NOTE:

$$\oint_{S_V} \vec{U} \cdot d\vec{S} = 0 \quad \forall S_V \in D \xRightarrow{\text{always}} \nabla \cdot \vec{U} = 0 \quad \forall p \in D$$

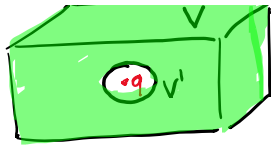
$$\nabla \cdot \vec{U} = 0 \quad \forall p \in D \xRightarrow{\uparrow} \oint_{S_V} \vec{U} \cdot d\vec{S} = 0 \quad \forall S_V \in D$$

ONLY if D is a SCD

Example: solid dielectric with VOID with charge



$$D = V - V' \quad (\text{MCD})$$



$$D = V - V' \quad (\text{MCD})$$

$$\rightarrow \nabla \cdot \vec{E} = \cancel{\rho/\epsilon} = 0 \Rightarrow \vec{E} \text{ is DIVERGENCE-FREE}$$

$$\forall p \in D$$

~~$\vec{D}$ is NOT SCD~~

$\vec{E}$ is a SOLENOIDAL FIELD

$$\text{CURL } \mathbb{R}^3 \Rightarrow \mathbb{R}^3$$

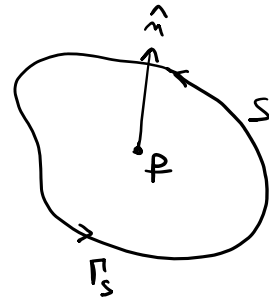
tendency of a V.F. to CIRCULATE around a point

let $\vec{U} \in C_1$ VF on D

$$\text{Def: } \nabla \times \vec{U}(p) \cdot \hat{m} = \lim_{\Delta S \rightarrow 0} \frac{\oint_{\Gamma_S} \vec{U} \cdot d\vec{l}}{\Delta S} =$$

area of S $\left[\frac{\text{CIRCULATION}}{\text{AREA}} \right]$

local circulation density

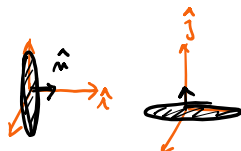


Component of curl of $\vec{U}$ along $\hat{m}$ = circulation per unit of surface of $\vec{U}$ along the boundary of an infinitesimal loop around p

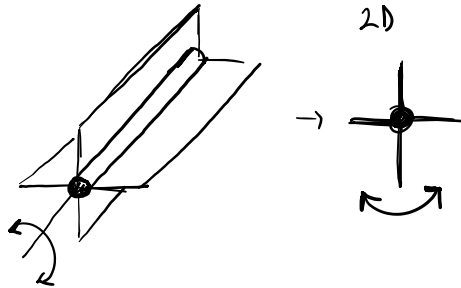
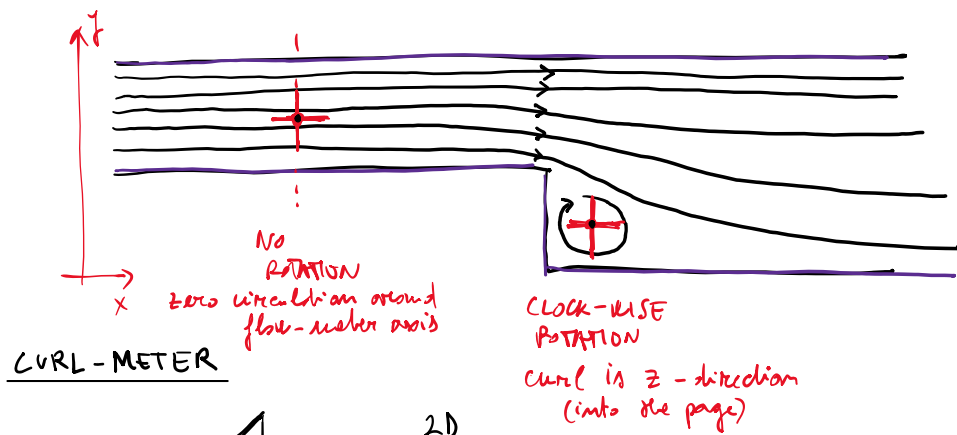
$$\text{CARTESIAN COORDINATES: } \nabla \times \vec{U}(p) = \text{DET} \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ U_x(p) & U_y(p) & U_z(p) \end{bmatrix}$$

$$= \hat{i} \left(\frac{\partial U_z(p)}{\partial y} - \frac{\partial U_y(p)}{\partial z} \right) - \hat{j} \left(\frac{\partial U_z(p)}{\partial x} - \frac{\partial U_x(p)}{\partial z} \right) + \hat{k} \left(\frac{\partial U_y(p)}{\partial x} - \frac{\partial U_x(p)}{\partial y} \right)$$

COMPONENT along $\hat{i}$ of CURL



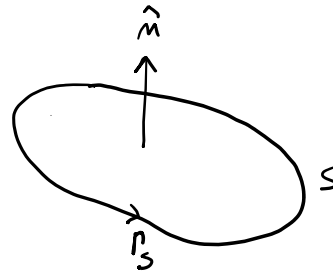
Example: turbulent flow in a duct



STOKES theorem

let $\vec{U} \in C^1$ on D

$$\underbrace{\int_S \nabla \times \vec{U} \cdot d\vec{S}}_{\text{flux of curl of } \vec{U} \text{ through } S} = \underbrace{\oint_{\Gamma_S} \vec{U} \cdot d\vec{\ell}}_{\text{circulation of } \vec{U} \text{ along boundary of } S}$$



IMPLICATION of STOKES theorem for CONSERVATIVE FIELDS

let $\vec{U}$ conservative field in $D \Leftrightarrow \oint_{\Gamma_S} \vec{U} \cdot d\vec{\ell} = 0 \quad \forall \Gamma_S \in D$

$$\underbrace{\oint_{\Gamma_S} \vec{U} \cdot d\vec{\ell}}_{=0} \stackrel{\text{STOKES}}{=} \underbrace{\int_S \nabla \times \vec{U} \cdot d\vec{S}}_{=0} \Rightarrow \boxed{\nabla \times \vec{U} = 0 \quad \forall \vec{p} \in D}$$

$\Uparrow$
 $\vec{U}$ is CURL-FREE on D

$$\begin{array}{ccc} \text{conservative field} & \text{always} & \text{curl-free} \\ \oint_{\Gamma_S} \vec{U} \cdot d\vec{\ell} = 0 \quad \forall \text{ closed } \Gamma_S & \Rightarrow & \nabla \times \vec{U} = 0 \quad \forall \vec{p} \in D \end{array}$$

curl-free $\implies$ conservative
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only if D is SCD